Basic ARM64 Assembly Guide

Register	Syntax	Purpose
Link register	`lr` or `x30`	holds return address
Stack pointer	`sp`	holds base address of current stack frame
Register 0		General purpose, input arguments, return values
	`x0`	all 64 bits
	`w0`	last 32 bits (first 32 bits set to 0)
Register 1		General purpose, input arguments
(also $2, \ldots, 7$)	`x1`	all 64 bits
	`w1`	last 32 bits (first 32 bits set to 0)
Register 8		General purpose
(also $9, \ldots, 15$)	`x8`	all 64 bits
	`w8`	last 32 bits (first 32 bits set to 0)

Instruction	What it does
`str x, [Mem]`	Store `x`'s value into memory at address Mem
`str x, [Mem, a]`	Store `x`'s value into memory at address Mem + a
`ldr x, [Mem]`	Load data stored at Mem into register x
`ldr x, [Mem, a]`	Load data stored at Mem + a into register x
`mov x, y`	Copy y's value into `x`
`sub x, y, z`	$x = y - z$
`add x, y, z`	$x = y + z$
`cmp x, y`	compare x to y, set conditions
`b LABEL`	go to LABEL (unconditionally)
`bl LABEL`	call procedure at LABEL
`blt LABEL`	go to LABEL if condition says $x < y$
`ble LABEL`	go to LABEL if condition says $x \leq y$
`bgt LABEL`	go to LABEL if condition says $x > y$
`bge LABEL`	go to LABEL if condition says $x \geq y$
`beq LABEL`	go to LABEL if condition says $x = y$
`bne LABEL`	go to LABEL if condition says $x \neq y$
`ret`	branch unconditionally to the address in `x30` = `lr`